CSE 3967: MACHINE LEARNING CONCEPTS 1

Sample Questions

1. a) State the difference between classical programming and Machine Learning in terms of how logic is obtained and how output is produced. Explain how learning from historical data supports prediction, using one real-world example.

   b) Given data = np.array([[10, -20, 30], [40, -50, 60], [70, 80, -90]]), write NumPy statements to calculate the mean and maximum of each column.

   c) Write the NumPy syntax to generate 50 samples from a normal distribution and 50 samples from a Uniform distribution. Explain the parameters and the role of a random seed.
2. A DataFrame named as df contains columns such as Name, Marks and Age. Marks contains [80, None, 70, None] and Age contains [20, 21, None, 22].

   a) Write code to count null values in every column.

   b) Replace missing Marks values with the mean of the non-missing Marks values.

   c) Create a second DataFrame containing only complete rows and explain the trade-off between deletion and imputation.
3. Daily customer counts for a shop are [42, 35, 51, 48, 35, 44, 60].

   a) Calculate the mean, median. Identify the mode, if any.

   b) A new observation of 100 is added. Explain which measure of central tendency is affected more by the newly added observation and why?

   c) Explain what standard deviation tells us about dispersion.
4. The electricity consumption from Jan to Jun is [210, 230, 195, 250, 280, 265] .

   a) Write Python code to draw a line plot and a bar chart with titles and axis labels.

   b) Annotate each bar with its consumption value.

   c) Write code to create a histogram and explain what additional information it provides compared with the line plot.
5. A DataFrame named as df has columns **acc**, **time** and **distance**, where distance is the target.

   a) Write code to calculate and display correlations between each input feature and distance.

   b) Write code to visualize distance against acc and time in a 1×2 grid of plots.

   c) Explain why a strong correlation does not, by itself, prove guarantee predictive usefulness.
6. A real-estate company has a .CSV dataset with area_sqft, bedrooms, bathrooms, location and price.

a) Write a Python workflow to load the data, inspect its shape, check missing values, and select area_sqft as X and price as y.

b) Split the data into 80% training and 20% testing subsets, train LinearRegression, and predict prices for the test set.

c) Explain why the test set should not be used to fit the model.

7. a) For y_actual = [10, 20, 30, 40, 50] and y_pred = [12, 18, 29, 42, 48], calculate MAE, MSE and RMSE.

   b) Show the error for each observation and the formula used.

   c) Write Scikit-learn code to calculate MAE, MSE, RMSE and R^2. Explain which of MAE, MSE and RMSE penalizes large errors more strongly, and interpret what an R^2 close to 1 indicates.

8. a) Given the relationship $y = 2 + 3x + 0.5x^2$, calculate y for x = 1, 2, 3 and 4. Explain why a linear regression estimator can be used with polynomial features.

   b) Write Scikit-learn code to generate polynomial features of degree 2 and fit a LinearRegression model.

   c) Explain how model complexity can affect fit and generalization.

9. a) A weather dataset contains Temperature and Humidity. Write a python code to create a feature, Statistical_Mean, equal to the average of these two features for each sample.

   b) Write code to add the feature.

   c) For a DataFrame containing acc, time and distance, write code to rank input features by absolute correlation with distance. Explain one benefit and one risk of adding engineered features.

10. a) Consider: X = df["area"]; y = df["price"]; X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42); model = LinearRegression(); model.fit(X_train, y_train). The fit call raises an input-shape error. Identify the likely cause of the error.

    b) Write the corrected statement for X and explain the expected shape of a single-feature input matrix.

    c) State two checks that should be performed before model fitting.

11. A predictive model continues to make errors even after its parameters have been optimized.

    a) Distinguish between errors caused by sub-optimal parameter values, noisy data, omitted features, and a model that cannot adequately represent the underlying relationship.

    b) Illustrate your answer with a suitable example and suggest practical approaches to investigate and reduce these errors.

c) Explain the role of bias and variance in model prediction errors. How can you determine whether a model is underfitting or overfitting by comparing its training and test errors?

12. a) Explain the bias–variance trade-off in machine learning with the help of a suitable graph. Discuss the relationship between model complexity, bias, variance, and generalization error.

b) Three regression models (A, B, and C) are trained on the same dataset. Their training and test errors are reported in the following table.

Model	Training Error	Test Error
A	25	28
B	5	7
C	1	35

Based on the results, identify the models

a) that are most likely to suffer from **underfitting** and

b) that are most likely to suffer from **overfitting.**

c) Identify the model that demonstrates the best generalization performance with proper justification

13. a) Model A has training $R^2 = 0.62$ and testing $R^2 = 0.60$. Model B has training $R^2 = 0.99$ and testing $R^2 = 0.51$. Which model shows stronger evidence of overfitting? Justify your answer.

b) Explain underfitting and generalization in this context.

c) Give one possible intervention for Model B and explain why it may help.

14. a) A dataset contains 1000 samples. A train-test split uses test_size=0.2 and random_state=42. Calculate the number of training and testing samples.

b) Explain the purpose of random_state and why evaluation only on training data is unreliable.

c) Explain one limitation of a single train-test split and how repeated or fold-based validation can help.

15. a) Differentiate between regression and classification problems with suitable examples.

b) Describe how the class probabilities can be converted into class labels? Explain with examples.

c) State the purpose of the sigmoid function in Logistic Regression? Discuss the range of output values produced by the sigmoid function?

16. a) Classification is unsupervised learning. Is this statement true or false? justify your answer.

b) A healthcare organization aims to develop a machine learning model to classify individuals as *Anemic* or *Non-Anemic* based on their Hemoglobin level and Red Blood Cell (RBC) count. Identify the input features and target variable, and justify why this problem can be formulated as a supervised binary classification task.

c) Write python code to generate a synthetic dataset containing 200 uniformly sampled instances for each class using the following feature ranges:

- **Non-Anemic:** Hemoglobin: 13–17 g/dL; RBC count: 4.5–6.0 million cells/µL.
- **Anemic:** Hemoglobin: 7–11 g/dL; RBC count: 2.5–4.0 million cells/µL.

17.a) Discuss the working principle of Logistic Regression for binary classification with example.

b) Write the mathematical expression for the sigmoid function and state its output range. Explain how predicted probabilities are converted into class labels using a threshold of 0.5.

c) Explain why a sigmoid-based model is generally more appropriate than using an unconstrained linear-regression output as a probability.

18.a) The actual class labels are [1, 0, 1, 1, 0], and a logistic regression model produces the predicted probabilities [0.85, 0.30, 0.65, 0.40, 0.20]. Using a classification threshold of 0.5, convert the predicted probabilities into class labels.

b) Compare the predicted class labels with the actual class labels. Identify the correctly and incorrectly classified samples and calculate the classification accuracy.

c) Discuss how changing the classification threshold influences the class-label predictions of a logistic regression model.